

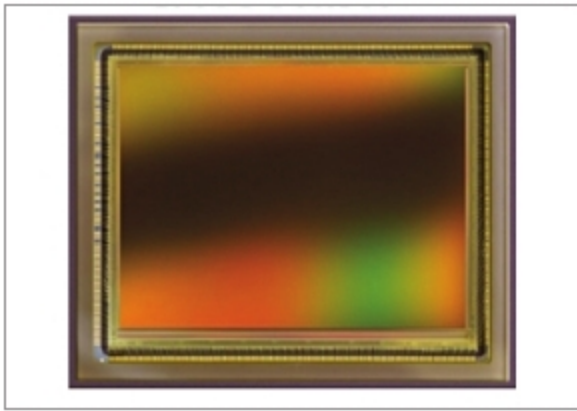


Fig. 11: CMOS sensor

extra circuit added to each pixel pushes out a single charge parallelly and generates an image.

In both CCD and CMOS cases, pixel information, in the form of light from the camera lens, is filtered through colour filters to disperse in red, blue, green, and white colours in the sensor.

- The more the light or photons, the brighter is the pixel. The brighter the pixel, the higher is the digital value. This digital value is described as a gray value.

- For a colour image, we need colour values for each individual pixel in an array. First, the array recognises the gray value (red, green, and blue) and calculates colour for each pixel from the gray value and then incorporates colour for the whole image.

Coming back to SMT technology, a picture is an image with an illustration of pixels. The quality of the image is usually expressed in microns.

The performance and capability of AOI depend on the choice of the light source, digital camera capability, speed of image acquisition, and data capture ability. The design and technology adopted with respect to camera specifications, illumination system, image capture, and processing differ from manufacturer to manufacturer.

Image processing system

The image processing system is purely based on the capability of the vision system and mechanism for extracting the information required in an application. In the majority of industrial applications, especially in AOI, the processing of images takes place externally through the PC base system. This process is performed by dedicated software with a wide variety of interfaces and tools.

This software performs several steps or algorithms. It acquires image from the sensor, preprocesses to opti-

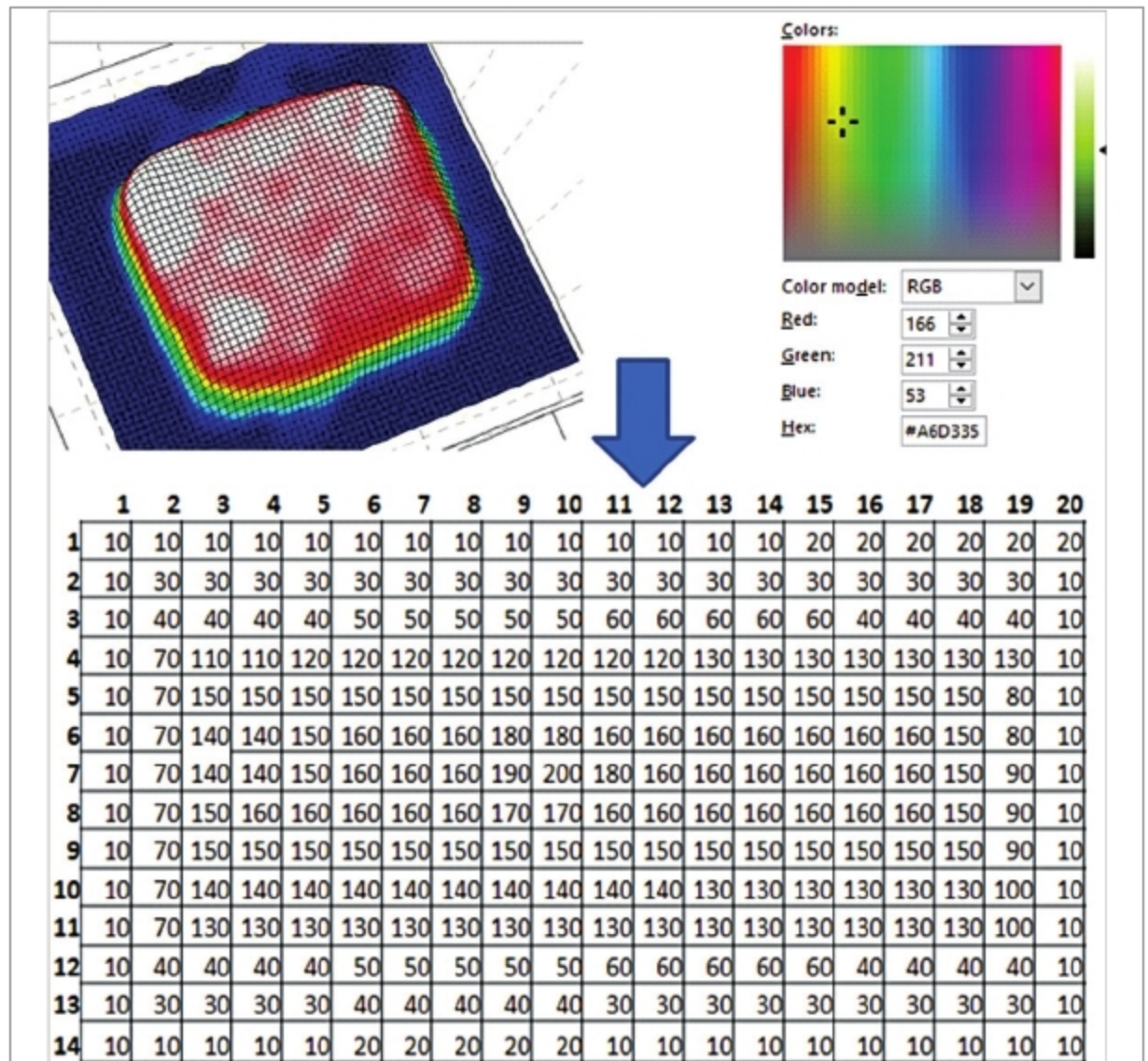


Fig. 12: In both CCD and CMOS sensors, pixel information is filtered through colour filters to disperse in red, blue, green, and white colours

mise the image, analyses according to features, sets the geometry and tolerance as per inspection conditions, and finally, results are communicated to the user in a defined format.

There are two types of technology-based AOI available in the market—Image-based AOI and algorithm-based AOI. The inspection process is different for both the technologies. Image-based AOI is quite popular, whereas algorithm-based AOI is preferred only in some applications and is quite expensive.

AOI programming

Creating a job file is the key to product programming. Here Kohyoung AOI machine is considered as an example to know the steps involved to create a job file. The creation of a job file is essential for AOI inspection as we need to assign detailed information such as package size and inspection area for every component. Here are the steps to create a job file for a target production PCB assembly model (Fig. 15).

The basic information needs to be provided to the AOI machine through Gerber data (image file) and CAD data

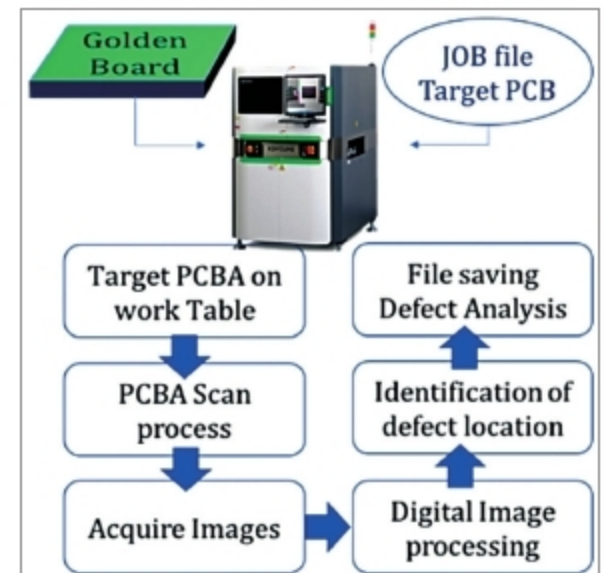


Fig. 13: Image based AOI inspection process

(placement data) to create a KY PCB file so that the machine can understand/read this PCB file. Machine EPM software collects the details about PCB size, masking area, footprints, arrays, and fiducial data. Gerber and CAD data is imported to create a PCB file as the first step of programming.

Target PCB is placed on conveyor to perform board level teaching—Fiducial, bad mark, and barcode. Once board-level teaching is defined, the images of the target PCB are grabbed through scanning. This scanning pro-